MILITARY SPECIFICATION

PROTECTORS, ELECTRICAL COMMUNICATION CIRCUITS

GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fused and fuseless protectors consisting of insulating base, arresters (air gap or gas filled), fuses and covers (when applicable), terminals for connection to line, instrument, ground and mounting facilities.

1.2 Classification.

1.2.1 Protector style. The protectors covered by this specification shall be of the following styles (see 3.1 and 6.2.2):

P1 1/ - Without cover, with arresters, type A1 or A4, and fuses, class F1.
P2 1/ - With metal or plastic cover, arresters, type A1 or A4, and fuses, class F2.
P3 - With plastic cover, arresters, type A2 or A3, fuseless.
P4 - With plastic cover, arresters, type A2 or A3, and fuses, class F2.

1.2.2 Arrester type. The arresters covered by this specification shall be of the following types (see 3.1 and 6.2.2):

A1 1/ - Carbon block (not fail safe) containing a discharge block and an insulating block containing a discharge electrode.
A2 - Carbon disc, fail safe (FS), providing permanent grounding of sustained high overcurrents by utilizing a fusible disc or spacer.
A3 - Gas tube, fail safe (FS), providing permanent grounding of sustained high overcurrents by utilizing a fusible disc, spacer, link or other suitable means.
A4 1/ - Gas tube (not fail safe) used primarily in protectors which include fuses.
A5 - Gas tube, 3 electrode, fail safe feature incorporated.

1.2.3 Fuse class. The fuses covered by this specification shall be of the following classes (see 3.1 and 6.2.2):

F1 1/ - Fibre or ceramic body with threaded terminals.
F2 - Ceramic body with dowel terminals.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

1/ Not for new design.

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SPECIFICATIONS

FEDERAL

FF-S-92	- Screws, Machine: Slotted, Cross-Recessed or Hexagon Head.
FF-W-92	- Washers, Metal, Flat (Plain).
QQ-A-250	- Aluminum Alloy Plate and Sheet; General Specification for.
QQ-B-613	- Brass, Leaded and Non-Leaded; Flat Products (Plate, Bar, Sheet, and Strip).
QQ-B-626	- Brass, Leaded and Non-Leaded; Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-B-750	- Bronze, Phosphor, Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections.
QQ-C-502	- Copper Rods and Shapes; and Flat Products with Finished Edges (Flat Wire, Strips, and Bars).
QQ-C-530	- Copper-Beryllium Alloy Bar, Rod and Wire (Copper Alloy Number 172).
QQ-C-533	- Copper-Beryllium Alloy Strip (Copper Alloy Numbers 170 and 172).
QQ-C-576	- Copper Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed or Machined Edges, (Plate, Bar, Sheet, and Strip).
QQ-N-290	- Nickel Plating (Electrodeposited).
QQ-S-763	- Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting.
QQ-S-766	- Steel Plates, Sheets, and Strip - Corrosion Resisting.
QQ-S-781	- Strapping, Steel, Flat and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.
PPP-B-676	- Boxes, Setup.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing).

MILITARY

MIL-I-10	- Insulating Compound, Electrical, Ceramic, Class L.
MIL-M-14	- Molded Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116	- Preservation, Methods of.
MIL-F-1148	- Fibre, Vulcanized, Electrical and Mechanical Grades.
MIL-T-10727	- Tin Plating, Electrodeposited or Hot-Dipped, for Ferrous and Nonferrous Metals.
MIL-F-14072	- Finishes for Ground Signal Equipment.
MIL-B-43014	- Boxes: Water Resistant Paperboard, Folding, Set-up and Metal-stayed.
MIL-C-45662	- Calibration System Requirements.
MIL-M-19590	- Marking of Commodities and Containers to Indicate Radio Active Material.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-147	- Palletized and Containerized Unit Loads 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1285	- Marking of Electrical and Electronic Parts.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services. (Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

PROTECTORS FOR COMMUNICATION CIRCUITS

Standard No. UL 497. (Application for copies should be addressed to Underwriters' Laboratories, Inc., 333 Pfingsten Road, Box 247, Northbrook, Illinois 60062.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein, and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.1.1 Detail requirements for protectors not covered by specification sheets. Unless otherwise specified, the protectors, arrestors, and fuses covered by the individual specification sheets are standard for military applications. This specification shall be applicable to nonstandard protectors, provided the desired requirements fall within the maximum and minimum requirements specified by an existing specification sheet. The qualified products list shall be applicable to these items.

3.2 Qualification. The protectors, arresters, and fuses furnished under this specification shall be products that conform to Underwriter's Laboratories Standard UL 497 which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.5 and 6.4). Qualification of a protector also qualifies a fuse or arrester associated with the protector.

3.3 First article. Unless otherwise specified (see 6.3), protectors furnished under this specification and not covered by specification sheets, or protectors which are not listed on the qualified products list shall be products which have passed the inspections specified in 4.6.

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3.4 Materials. The materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the protectors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4.1 Restricted material. Flammable or explosive material, or metal which can produce toxic or suffocating fumes when the protectors are in service shall not be used in the construction.

3.4.2 Fibre. Fibre shall conform to grade CH, form T, of MIL-F-1148.

3.4.3 Plastic material. Molded plastic material shall conform to MIL-M-14 or have Underwriters' Laboratories certification. Cotton base or cotton cellulose filled molding material shall not be used.

3.4.4 Ceramic. Ceramic material shall be as specified (see 3.1 and 6.2.2).

3.4.5 Metals. Metals shall be of a corrosion-resistant type, or shall be treated to resist corrosion.

3.4.5.1 Brass. Brass shall conform to QQ-B-613 or QQ-B-626.

3.4.5.2 Phosphor bronze. Phosphor bronze shall conform to QQ-B-750.

3.4.5.3 Aluminum. Aluminum shall conform to QQ-A-250.

3.4.5.4 Copper. Copper shall conform to QQ-C-502 and QQ-C-576.

3.4.5.5 Copper-beryllium-alloy. Copper-beryllium-alloy shall conform to QQ-C-530 or QQ-C-533.

3.4.5.6 Corrosion-resistant steel. Corrosion-resistant steel shall conform to QQ-S-763 or QQ-S-766 (hard or hardened).

3.4.5.7 Dissimilar metals. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metals in contact, which tend toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy), is not acceptable. However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. The use of dissimilar metals separated by a suitable insulating material is also permitted. Dissimilar metals are defined in 6.5 and table VI, except that the use of corrosion-resistant steel in contact with cadmium plating shall be allowed.

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3.5 Design and construction. The protectors shall be of the design, construction, and physical dimensions specified (see 3.1 and 6.2.2).

3.5.1 Threaded parts. Unless otherwise specified (see 3.1 and 6.2.2), all threaded parts shall be in accordance with Handbook H-28. Where practical, all threads shall be in conformity with the coarse-thread series. The fine-thread series shall be used only for applications that show a definite advantage through their use. Where a special diameter-pitch combination is required, the thread shall be of the Unified National form and of any pitch between 16 and 36 which is used in the firm-thread series. All threaded parts shall engage at least two full threads. Where a screw mates with a plastic part, a threaded metal insert shall be molded therein.

3.5.1.1 Nuts and screws. Unless otherwise specified (see 3.1 and 6.2.2), all nuts and screws shall conform to FF-S-92.

3.5.2 Brass washers. All brass washers shall conform to FF-W-92.

3.5.3 Finishing. All finishes shall conform to MIL-F-14072, MIL-T-10727, or QQ-N-290.

3.5.4 Securing of parts. Through-bolting shall be used when possible. Where practical, lock-washers shall be incorporated under assembly-securing screws. The current carrying parts shall be secured and ridigly fastened to a supporting or mounting base; a means of fastening may be accomplished by using screws, a connecting strap or clip adequately fitted into an adjacent part, or by some other equivalent method. The mechanical means used to secure parts to a supporting or mounting base, or to each other, shall be adequate to prevent loosening of the parts so secured when making wire connections.

3.5.5 Base. The base shall be of one-piece construction and the size, dimensions, and material shall be as specified (see 3.1, 3.4.3, and 6.2.2). Unless otherwise specified, wire entrance holes shall be provided with neoprene grommets to minimize the entrance of insects and moisture. Two entrances shall be provided on fused type protectors.

3.5.5.1 Current carrying parts. The current carrying parts shall in every case be separated from the grounded parts of the unit (including the cover, if metal) by a space of at least 3/8-inch. Current carrying parts shall be bronze, brass, carbon, or copper, except for gas tube electrodes (see 3.1).

3.5.6 Terminals. All terminals of protectors shall be as specified (see 3.1 and 6.2.2). The terminal studs shall be as specified (see 3.1 and 6.2.2), and supplied with plain brass washers and hexagon brass nuts.

3.5.7 Fuses (when applicable). Two fuses shall be provided with each unit and shall be as specified (see 3.1 and 6.2.2).

3.5.8 Fuseholders (contact springs) (when applicable). Fuseholder contact springs shall be phosphor bronze or beryllium copper and shall be designed with slots that accommodate No. 10 fuse terminal studs or .203-inch dowel terminals as specified (see 3.1 and 6.2.2). The contact springs metal shall be not less than 0.020 inch thick and 0.43 inch wide or equivalent cross-sectional area.

3.5.9 Arresters. Two arresters shall be provided with each protector and shall be as specified (see 3.1 and 6.2.2). Arresters of the same type (see 1.2.2) shall be physically and electrically interchangeable. Mica or acetate spacers (or other material with short leakage paths) shall not be used in the air gaps for a dielectric.

3.5.9.1 Arrester mounting method. The protector shall include suitable mounting for the specified arrester (see 3.1 and 6.2.2).

3.5.10 Covers. Each protector, unless otherwise specified, shall be provided with a removable weatherproof cover. If metallic, it shall be suitably treated to prevent corrosion. The material used for covers shall be as specified (see 3.1 and 6.2.2).

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3.6 Insulation resistance. When the protector is measured as specified in 4.8.2, the insulation resistance shall be not less than 500 megohms, unless otherwise specified.

3.7 Resistance to ground. When the arresters are tested as specified in 4.8.3.1, the resistance to ground shall be not less than 83,000 ohms. Any momentary or intermittent reading may be disregarded.

3.8 Voltage breakdown. The arresters shall have a breakdown voltage as specified (see 3.1) when tested in accordance with 4.8.3.2.

3.9 Aging.

3.9.1 Accelerated air oven aging. When the protectors are tested as specified in 4.8.4.1, there shall be no evidence of mechanical or electrical damage. Melting of the fusible component in fail safe arrester assemblies shall not constitute a failure.

3.9.2 Ultraviolet light aging. When the protectors are tested as specified in 4.8.4.2, there shall be no evidence of mechanical or electrical damage.

3.9.3 Cold box aging. When the protectors are tested as specified in 4.8.4.3, there shall be no evidence of mechanical damage.

3.10 Vibration. When protectors are tested as specified in paragraph 4.8.5, there shall be no evidence of broken, loose, deformed, or misplaced parts.

3.11 Dielectric withstanding voltage. The protector shall be tested as specified in 4.8.6. There shall be no breakdown, flashover, arcing or current flow in excess of 1 milliampere.

3.12 Moisture resistance. When protectors are tested as specified in 4.8.7, the insulation resistance shall meet the requirements of 3.6.

3.13 Ability to resist permanent grounding. The arresters shall have the capability to resist permanent grounding when tested as specified in 4.8.8. Arresters shall be considered grounded when they fail to meet the requirements of 3.7. Following the test, insulation resistance shall meet the requirements of 3.6.

3.14 Fuses (when applicable).

3.14.1 Continuity. When tested as specified in 4.8.9.2, fuses shall have a maximum resistance of 0.2 ohms.

3.14.2 Current carrying capacity. When fuses are tested as specified in 4.8.9.3, current carrying capacity shall be 7 amperes continuous. During the test, the exterior of the fuse shall not exceed a rise in temperature of more than 70° C above ambient temperature.

3.14.3 Overload interrupt. Fuses shall interrupt 150-percent overload (10.5 amperes) in 5 minutes or less when tested as specified in 4.8.9.4. The circuit shall remain open without the circuit closing again during the 5-minute period after blowing. There shall be no mechanical failure and the insulation resistance shall be not less than 1/2 megohm; when tested in accordance with 4.8.2.1.

3.14.4 Short circuit interrupting capacity. The fuse shall be capable of interrupting a short circuit of 10,000 amperes at 600 volts without reclosing the circuit when tested as specified in 4.8.9.5. The fuses shall remain intact, the insulation shall not puncture, the terminals shall not separate from the fuse body. Fuses shall not emit sufficient flame or molten metal to ignite surgical cotton surrounding the case.

3.15 Salt spray (corrosion). When the protectors are tested as specified in 4.8.10, there shall be no evidence of excessive corrosion. Excessive corrosion is defined as that which interferes with the electrical or mechanical performance. There shall be no warping, cracking, or other damage to the unit.

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3.16 Shock (specified pulse). When protectors are tested as specified in 4.8.11, there shall be no loose arrester or component parts nor shall there be any evidence of mechanical or electrical damage.

3.17 Thermal shock. When protectors are tested as specified in 4.8.12, there shall be no mechanical or electrical damage, or loosening of fastening devices.

3.18 High temperature. When the protectors are tested as specified in 4.8.13, covers shall be capable of withstanding the stress of removal and reinstallation. There shall be no evidence of mechanical or electrical damage.

3.19 Low temperature. When protectors are tested as specified in 4.8.14, covers shall be capable of withstanding the stress of removal and reinstallation. There shall be no evidence of mechanical or electrical damage.

3.20 Terminal strength. Protectors and fuses shall withstand the applied force without movement or discernible damage when tested as specified in 4.8.15.

3.21 Interchangeability. Replaceable protective parts (fuses and arresters) of any style protector shall be mechanically and electrically interchangeable with corresponding parts of any other protector of the same style.

3.22 Marking. Unless otherwise specified (see 3.1 and 6.2.2), protectors shall be marked in accordance with MIL-STD-130 and the following:

(a) Part number (see 3.1).
(b) Manufacturer's name or trademark, date and source code shall be in accordance with MIL-STD-1285.
(c) Arresters and protectors containing intentionally added radio active materials shall be marked as specified in MIL-M-19590.

3.22.1 Terminal identification. Ground terminals shall be designated by the letter "G". Line terminals if applicable shall be designated by the letter "L". Instrument terminals shall be designated by the letter "I" unless instrument and line terminals are coincident.

3.23 Workmanship. Protectors shall be processed in such a manner as to be uniform in quality, and the interiors of the protectors shall be free from metal filings, grease or oil, foreign material, dust or other loose particles that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).
(b) Qualification inspection (see 4.5).
(c) First article inspection (see 4.6).
(d) Quality conformance inspection (see 4.7).

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4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table I, used in fabricating the protectors, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE I. Materials inspection.

Material	Requirement paragraph	Applicable specification
Fibre	3.4.2	MIL-F-1148
Plastic	3.4.3	MIL-M-14
Ceramic	3.4.4	MIL-I-10
Metals	3.4.5	MIL-F-14072
Brass	3.4.5.1	QQ-B-613, QQ-B-626
Phosphor bronze	3.4.5.2	QQ-B-750
Aluminum	3.4.5.3	QQ-A-250
Copper	3.4.5.4	QQ-C-502, QQ-C-576
Copper-beryllium alloy	3.4.5.5	QQ-C-530, QQ-C-533
Corrosion-resisting steel	3.4.5.6	QQ-S-763, QQ-S-766
Finishes	3.5.3	MIL-F-14072
Tin plate	3.5.3	MIL-T-10727
Nickel plate	3.5.3	QQ-N-290

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.4) on sample units produced with equipment and procedures normally used in production.

4.5.1 Sample size. Eight protectors shall be subjected to qualification inspection.

4.5.2 Inspection routine. The sample shall be subjected to the inspections specified in table II, in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall then be divided equally into two groups of four units each.

4.5.3 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.5.4 Retention of qualification. To retain qualification, the supplier shall forward a report at 24-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery, (groups A and B), indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.

(b) A summary of the results of tests performed for qualification verification inspection, (group C), including the number and mode of failures. The summary shall include results of all qualification verification inspection tests performed and completed during the 24-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list. Failure to submit the report within 90 days after the end of each 24-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time during the 24-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification. In the event that no production occurred during the reporting period, a report shall be submitted

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TABLE II. Qualification inspection.

Examination or test	Requirement paragraph	Method paragraph
Group I (all sample units)		
Visual & mechanical examination	3.1 thru 3.5.10, 3.22 and 3.23	4.8.1
Insulation resistance	3.6	4.8.2.1
Resistance to ground	3.7	4.8.3.1
Voltage breakdown	3.8	4.8.3.2
	3.6.3	
Group II (four sample units)	3.7	
Aging	3.9	4.8.4
Accelerated air oven aging	3.9.1	4.8.4.1
Ultraviolet light aging	3.9.2	4.8.4.2
Cold box aging	3.9.3	4.8.4.3
Vibration	3.10	4.8.5
Dielectric withstanding voltage	3.11	4.8.6
Moisture resistance	3.12	4.8.7
Ability to resist permanent grounding	3.13	4.8.8
Fuses (when applicable)	3.14	4.8.9
Terminal strength	3.20	4.8.15
Continuity	3.14.1	4.8.9.2
Current carrying capacity	3.14.2	4.8.9.3
Overload interrupt	3.14.3	4.8.9.4
Short circuit interrupting capacity	3.14.4	4.8.9.5
Group III (four sample units)		
Salt spray (corrosion)	3.15	4.8.10
Shock (specified pulse)	3.16	4.8.11
Thermal shock	3.17	4.8.12
Moisture resistance	3.12	4.8.7
High temperature	3.18	4.8.13
Low temperature	3.19	4.8.14
Terminal strength	3.20	4.8.15
Resistance to ground	3.7	4.8.3.1
Voltage breakdown	3.8	4.8.3.2
Ability to resist permanent grounding	3.13	4.8.8

certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product of each type or style, to testing in accordance with the qualification inspection requirements.

4.6 First article inspection. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders. First article inspection shall be identical to that required for qualification.

4.7 Quality conformance inspection.

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4.7.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.7.2 Inspection lot. An inspection lot shall consist of all protectors or components of the same style or type, produced under essentially the same conditions, and offered for inspection at one time.

4.7.2.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table III, in the order shown.

4.7.2.1.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105. Only two sample units shall be subjected to verification of dimensional requirements.

4.7.2.1.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE III. Group A inspection.

Examination or test	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination	3.1 thru 3.5.10, 3.22 and 3.23	4.8.1	1.0	4.0
Insulation resistance	3.6	4.8.2.1	1.0	---
Resistance to ground	3.7	4.8.3.1		
Voltage breakdown	3.8	4.8.3.2		

4.7.2.2 Group B inspection. Group B inspection shall consist of the tests specified in table IV, in the order shown, and the sample shall be selected from inspection lots that have passed group A inspection.

4.7.2.2.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-1. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 4 percent defective.

4.7.2.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE IV. Group B inspection.

Examination or test	Requirement paragraph	Test method paragraph
Terminal strength	3.20	4.8.15
Dielectric withstanding voltage	3.11	4.8.6
Fuses (when applicable)		
Terminal strength	3.20	4.8.15
Continuity	3.14.1	4.8.9.2
Current carrying capacity	3.14.2	4.8.9.3
Overload interrupt	3.14.3	4.8.9.4

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4.7.2.2.3 Disposition of sample units. Sample units subjected to group B inspection may be delivered on the contract or purchase order, if the lot is accepted. These fuses must be replaced from lots which have passed group A inspection before the lot is delivered on the contract or purchase order.

4.7.3 Qualification verification inspection. Qualification verification inspection shall consist of group C. Except where the results of these inspection show noncompliance with the applicable requirements (see 4.7.3.1.4), delivery of products which have passed groups A and B shall not be delayed pending the results of these qualification verification inspections.

4.7.3.1 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table V, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection.

4.7.3.1.1 Sampling plan. Eight sample units shall be selected from those covered by a single style or type during the 24-month period following the date of notification of qualification, and during each subsequent 24-month period.

4.7.3.1.2 Failures. If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed.

TABLE V. Group C inspection.

Examination or test	Requirement paragraph	Test method paragraph	Number of sample units to be inspected
Subgroup I			
Insulation resistance	3.6	4.8.2.1	
Resistance to ground	3.7	4.8.3.1	
Voltage breakdown	3.8	4.8.3.2	
Vibration	3.10	4.8.5	4
Dielectric withstanding voltage	3.11	4.8.6	
Moisture resistance	3.12	4.8.7	
Ability to resist permanent grounding	3.13	4.8.8	
Fuses (when applicable)	3.14	4.8.9	
Short circuit interrupting capacity	3.14.4	4.8.9.5	
Aging (when applicable) 1/			
Accelerated air oven aging	3.9.1	4.8.4.1	
Ultraviolet light aging	3.9.2	4.8.4.2	
Cold box aging	3.9.3	4.8.4.3	
Subgroup II			
Salt spray (corrosion)	3.15	4.8.10	
Shock (specified pulse)	3.16	4.8.11	
Thermal shock	3.17	4.8.12	
Moisture resistance	3.12	4.8.7	
High temperature	3.18	4.8.13	
Low temperature	3.19	4.8.14	4
Vibration	3.10	4.8.5	
Terminal strength	3.20	4.8.15	
Resistance to ground	3.7	4.8.3.1	
Voltage breakdown	3.8	4.8.3.2	
Ability to resist permanent grounding	3.13	4.8.8	

1/ If it can be verified that the same type and grade of plastic is being used and that essentially the same fabrication processes are being utilized as was used at time of qualification, the aging tests need not be performed. Certification shall be supplied.

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4.7.3.1.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.7.3.1.4 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection,, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.7.4 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4.8 Methods of examination and test.

4.8.1 Visual and mechanical examination. Protectors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the requirements specified herein (see 3.1 thru 3.5.10, 3.22 and 3.23).

4.8.2 Insulation resistance (see 3.6).

4.8.2.1 Insulation resistance (without arresters). The protectors shall be measured in accordance with method 302, test condition B, of MIL-STD-202. The points of measurement shall be between each terminal to ground, and between terminals.

4.8.2.2 Insulation resistance (with arresters). The protectors shall be measured in accordance with method 302, test condition A, of MIL-STD-202. The points of measurement shall be between each terminal to ground, and between terminals.

4.8.3 Voltage tests.

4.8.3.1 Resistance to ground (see 3.7). Arresters shall be tested for grounding by the following method: Connect a 100,000-ohm resistor in series with the arrester and determine the resistance by applying a potential of 110 Vdc to the terminals of the combination and measuring the resultant circuit current and the drop in potential across the arrester.

4.8.3.2 Voltage breakdown (see 3.8). The arresters which successfully pass the test of 4.8.3.1 shall be subjected to the following breakdown test. The arresters shall be subjected to a potential below the dc breakdown voltage (see 3.1). The voltage shall then be increased at a rate less than 100 volts per second until breakdown occurs. This breakdown shall be as specified (see 3.1 and 6.2.2). The magnitude of the power supply shall be limited to a maximum of 2.0 amperes and 300 volts over the specified breakdown voltage rating (see 3.1 and 6.2.2). A circuit opening device shall be employed to limit the time of discharge to 0.5 second, maximum. The initial firing voltage may be ignored.

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4.8.4 Aging (see 3.9).

4.8.4.1 Accelerated air-oven aging. Protectors shall be subjected to an accelerated aging test consisting of 96 hours in a circulating air oven maintained at not less than 100° C. After the aging period, the protectors shall be examined for signs of deterioration as a result of air-oven aging.

4.8.4.2 Ultraviolet light aging.

(a) Protectors shall be exposed to an ultraviolet light from a single enclosed violet carbon arc, using the apparatus described below. Water shall be automatically sprayed onto the sample units at predetermined intervals.

(b) The arc shall be formed between the upper electrode and lower vertical electrodes held in a solenoid-actuated feed mechanism. All electrodes shall be of carbon 1/2-inch in diameter, the upper electrode being of the solid type and the lower electrodes being of the neutral cored type, or vice versa. The potential across the arc shall be 120 to 125 Vac and the operating current shall be 15 to 17 amperes. The arc shall be enclosed by a clear globe of Pyrex 9200EX glass.

(c) The protectors shall be mounted on the interior of a cylindrical drum facing the arc, this cylinder being rotated about the arc at 1 rpm. The drum shall have an effective inside diameter of 30 inches and shall be 17-3/4 inches in height.

(d) Water, at room temperature and about 12 lbf/in^2 pressure, shall be sprayed horizontally onto the protectors through an assembly of nozzles. The flow of water and the duration of the spraying shall be controlled by a solenoid value operating in conjunction with a motor-driven timer. A chamber, having a partially recirculated mechanical ventilation and a temperature control, shall surround the arc, the revolving cylinder in which the protectors are mounted, and the water spray nozzles. The temperature of the protectors shall be measured by a block panel assembly consisting of a dial type thermometer with a sensing element fastened to a sheet of stainless steel. The sheet and the thermometer element shall be coated with black enamel. This black enamel panel assembly shall be mounted on the interior of the revolving drum, at the same distance from the arc as the protectors.

(e) During each operating cycle of the apparatus (20 minutes) each protector shall be exposed to ultraviolet light from the carbon arc for 17 minutes, and the ultraviolet light and water spray for 3 minutes. The test shall be continued until the cover has been exposed to the ultraviolet light for a total of 36 hours. The temperature within the apparatus as measured by the black panel assembly shall be 60° C (140° F). The covers shall show no evidence of undue aging or deterioration as a result of the exposure to ultraviolet light.

4.8.4.3 Cold box aging. Protectors shall be placed in a cold chamber for a period of 96 hours at a temperature of -29° C or colder. Immediately upon removal from the cold chamber, three sample covers shall be dropped to an asphalt tile floor from a height of 5 feet and examined for damage. The protectors shall then be allowed to warm to room temperature and then be examined for signs of deterioration, cracking, or breakage.

4.8.5 Vibration (see 3.10). Protectors shall be tested in accordance with method 204, test condition A, of MIL-STD-202. The following details shall apply:

(a) Units shall be rigidly mounted by normal means.

(b) Dislodging of test fuse during test shall be considered a failure.

(c) Units shall be examined for loose, broken, or cracked components.

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4.8.6 Dielectric withstanding voltage (see 3.11). Protectors shall be tested in accordance with method 301 of MIL-STD-202. (Arresters shall be removed.) The following details shall apply:

(a) Method of mounting - Normal.

(b) Test voltage - 1,000 V.

(c) Nature of potential - Alternating current.

(d) Points of application - Between each terminal to ground, and between terminals.

4.8.7 Moisture resistance (see 3.12). Protectors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - Normal.

(b) Polarizing voltage on half the specimens - 100 Vdc.

(c) Cold and vibration, steps 7a and 7b, shall be omitted.

(d) Measurements after test - Insulation resistance shall be measured as specified in 4.8.2.1 immediately after the 24-hour drying period.

4.8.8 Ability to resist permanent grounding (see 3.13). The arresters shall remain operative when a 2-microfarad capacitor, charged to not more than 300 volts over the specified dc breakdown voltage (see 3.1) is applied to the arrester with a 0.5 ohm resistor in series. The number of discharges shall be as specified (see 3.1). An interval of 2 minutes or more shall be permitted between successive discharges. Following the test, the resistance to ground shall be tested as specified in 4.8.3.1.

4.8.9 Fuses (see 3.14).

4.8.9.1 Special conditions. The fuse shall be mounted in a suitable protector and the protector shall be mounted in the normal mounted position for the tests specified in 4.8.9.2, 4.8.9.4, and 4.8.9.5. Arresters shall be removed to prevent arcing and/or other damage. Test leads shall be wire size AWG14 and the length kept to a minimum.

4.8.9.2 Continuity. Continuity of the fuse shall be determined by using a suitable ohmeter. The points of connection shall be the instrument and line terminals of the protector.

4.8.9.3 Current carrying capacity. Fuses shall be subjected to an alternating or direct current of 7 amperes. Current shall be maintained for not less than 30 minutes after temperature has stabilized. It may be assumed that the temperature has stabilized when three successive temperature readings, taken at 10-minute intervals, indicate no increase in temperature (within $\pm 1^{\circ}$ C). Thermocouple leads shall be wire size AWG30.

4.8.9.4 Overload interrupt. The fuses shall be maintained at the test temperature (see 4.8.9.1) for a minimum of 30 minutes prior to applying the overload current of 150 percent (10.5 amperes). The power supply shall have an open circuit voltage of not less than 600 volts. Following overload interrupt test, insulation resistance shall be tested as specified in 4.8.2.1.

4.8.9.5 Short circuit interrupting capacity (see 3.14.4). Fuses shall be subjected to the current and voltage as specified (see 3.14.4). DC tests shall be made using appropriate generating equipment with a rate of current rise for the test circuit adjusted for at least 3.25×10^6 amperes per second. AC tests shall be conducted using an ac supply of adequate capacity and shall be on a single phase basis. The current shall be applied within $\pm 10^{\circ}$ of zero point of the voltage wave. Circuit power factor shall be 0.9 or greater. Short circuit currents shall be determined by means of an oscillograph. Test circuits shall be calibrated for the specified current with the applicable fuseholder (protector) short circuited.

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4.8.10 Salt spray (corrosion) (see 3.15). Protectors shall be tested (with covers in place) in accordance with method 101 of MIL-STD-202. The following details shall apply:

(a) Test condition - A.

(b) Examination after exposure - The unit shall be examined for corrosion or damage after testing.

4.8.11 Shock (specified pulse) (see 3.16). Protectors shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

(a) Mounting method - Normal means.

(b) Test condition - G.

(c) Examination after shock - Protectors shall be examined for evidence of mechanical or electrical damage.

4.8.12 Thermal shock (see 3.17). Protectors shall be subjected to thermal shock in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition - A.

(b) Measurements after cycling - The tested units shall be examined for loose, broken, or cracked components.

4.8.13 High temperature (see 3.18). The units shall be maintained for 24 hours in a chamber at an ambient temperature of 85° C. The temperature shall then be lowered to 55° C and stabilized for 1 hour and the cover shall then be removed and reinstalled.

4.8.14 Low temperature (see 3.19). The units shall be maintained for 24 hours in a chamber at an ambient temperature of -65° C. The temperature shall be raised to -40° C and stabilized for 1 hour, and the cover shall then be removed and reinstalled.

4.8.15 Terminal strength (see 3.20). The terminals shall be subjected to the torque tests in accordance with test condition E, method 211 of MIL-STD-202.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Protectors and component parts shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Protectors and component parts shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Protectors and component parts (arrestor assemblies and fuses) shall be individually packaged in accordance with the methods of MIL-P-116 designated herein insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein.

5.1.1.4.1 Protectors. Protectors shall be packaged in accordance with submethod IC-2. The container shall conform to PPP-B-566 or PPP-B-676.

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5.1.1.4.2 Arrestor assemblies.

5.1.1.4.2.1 Carbon block arrestor assemblies. Carbon block arrestor assemblies shall be packaged in accordance with method III.

5.1.1.4.2.2 Carbon disc and gas tube arrestor assemblies. Carbon disc and gas tube arrestor assemblies shall be packaged in accordance with submethod IC-1.

5.1.1.4.3 Fuses. Fuses shall be packaged in accordance with method III.

5.1.1.5 Intermediate packaging. Arrestor assemblies and fuses, packaged as described in 5.1.1.4.2 and 5.1.1.4.3, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape and quantities, shall be of minimum tare and cube and shall contain identical items in multiples of five unit packages, not to exceed 50 unit packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 unit packages. Intermediate packaging is not required for protectors.

5.1.2 Level C. Protectors and component parts shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The packaged protectors and component parts shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using non-metallic or tape banding only.

5.2.2 Level B. The packaged protectors and component parts shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged protectors and component parts shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Protectors and component parts, packed as specified in 5.2.1, shall be placed on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Protectors and component parts, packed as specified in 5.2.2, shall be palletized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Protectors and component parts, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

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5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Navy procurements. For Navy procurements the use of polystyrene loose fill material (such as strips, strands and beads) is prohibited for packaging and packing applications.

5.4.3 Army procurements. (See 5.1.1.4.2.1, 5.1.1.4.3, 5.1.1.5, 5.2.1, 5.2.2 and 5.2.4.2.) All unit and intermediate containers shall either be weather resistant or overwrapped with waterproof barrier materials. Containers conforming to PPP-B-566 or PPP-B-676 shall be overwrapped with waterproof barrier materials or shall conform to MIL-B-43014. For level A packing when quantities per destination are less than a unitized load, the fiberboard containers shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with the applicable container specification except that metal strapping shall conform to QQ-S-781, type I, class B. For level B packing fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded. Weather resistant fiberboard caps shall be used for level B unitized loads.

6. NOTES

6.1 Intended use. Protectors covered by this specification are intended for use primarily in ground communication circuits for limiting lightning surges and other excessive currents.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Protectors covered by specification sheets.

(a) Title, number, and date of this specification.

(b) Title, number, and date of the applicable specification sheet and military part number (see 3.1).

(c) Level of preservation and packaging and applicable marking (see section 5).

6.2.2 Protectors not covered by specification sheets.

(a) Title, number, and date of this specification.

(b) Protector style, arrester type, and fuse class.

(c) Information pertaining to inspection (see 3.3 and 4.6):
(1) The laboratory at which the inspection is to be performed (see 4.6).
(2) Submission of samples (see 4.6).

(d) Applicable design and construction requirements (see 3.5 through 3.5.10).

(e) Marking, if other than specified in 3.22.

(f) Voltage breakdown (see 4.8.4.2).

(g) Ability to resist permanent grounding (see 4.8.8) - specify number of discharges.

(h) Levels of preservation and packaging and packing, and applicable marking (see section 5).

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6.3 First article inspection. Information pertaining to first article inspection of products covered by this specification should be obtained from the procuring activity for the specific contracts involved.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the U. S. Army Electronics Command, AMSEL-PP-EM, Fort Monmouth, NJ 07703; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), Dayton, Ohio 45401, agent for administration of the qualified products list.

6.4.1 Application for qualification. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" which may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.5 Dissimilar metals (see 3.4.5.7).

6.5.1 Intermetallic contact. The finishing of metallic areas to be placed in intimate contact by assembly presents a special problem, sinch intermetallic contact of dissimilar metals results in electrolytic couples which promote corrosion through galvanic action. To provide the required corrosion protection, intermetallic couples are restricted to those permitted by table VI. Table VI shows metals and alloys (or plates) by groups which have common electromotive forces (EMF) within 0.05 volt when coupled with a saturated calomel electrode in sea-water at room ambient temperatures. All members of a group are considered as completely compatible, one with the other. Compatible couples between groups have been specified in table VI based on a potential difference of 0.25 volt maximum. To simplify any arithmetic involved, table VI shows, in addition to EMF, against a calomel electrode, a derived "anodic index" with group 1 (gold, etc.) as 0 and group 18 (magnesium, etc.) as 175. Subtraction of a lower group anodic index gives the EMF difference in hundredths of a volt.

6.5.2 Groups. Table VI sets up 18 primary groups. It may be noted that neither metallurgical similarity or dissimilarity of metals is the parameter for selection of compatible couples. All members within a group, regardless of metallurgical similarity, are considered inherently nonsusceptible to galvanic action when coupled with any member within the group; for example, such dissimilar metals as platinum and gold. Similarly, such basically dissimilar alloys as austenitic. stainless steel, silver-solder, and low brass (all members of group 5) are inherently nonsusceptible when coupled together.

6.5.3 Compatibility graphs. Permissible couple series are shown in table VI by the graphs at the right. Members of groups connected by lines will form permissible couples. A 0 indicates the most cathodic member of each series, a ● an anodic member, and the arrow indicates the anodic direction.

6.5.4 Selection of compatible couples. Proper selection of metals in the design of equipment with result in fewer intermetallic contact problems. For example, for sheltered exposure, neither silver nor tin require protective finishes. However, since silver has an anodic index of 15 and tin 65, the EMF generated as a couple is 0.50 volt which is not allowable by table VI. In this case, other metals or plates will be required. It should be noted that, in intermetallic couples, the member with the higher anodic index is anodic to the member with the lower anodic index and will be susceptible to corrosion in the presence of an electrolytic medium. If the surface area of the cathodic part is significantly greater than that of the anodic part, the corrosive attack on the contact area of the anodic part may be greatly intensified. Material selection for intermetallic contact parts, therefore, should establish the small part as the cathodic member of the couple, whenever practicable.

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TABLE VI. Compatible couples (see 6.5) 1/

Group No.	Metallurgical category	EMF (volt)	Anodic index (0.01 v)	Compatible couples
1	Gold, solid and plated; gold-platinum alloys; wrought platinum (most cathodic)	+0.15	0	
2	Rhodium plated on silver-plated copper	+0.05	10	
3	Silver, solid or plated; high silver alloys	0	15	
4	Nickel, solid or plated; monel metal, high nickel-copper alloys	-0.15	30	
5	Copper, solid or plated; low brasses or bronzes; silver solder; German silver; high copper-nickel alloys; nickel-chromium alloys; austenitic corrosion-resistant steels	-0.20	35	
6	Commercial yellow brasses and bronzes	-0.25	40	
7	High brasses and bronzes; naval brass; Muntz metal	-0.30	45	
8	18 percent chromium type corrosion-resistant steels	-0.35	50	
9	Chromium, plated; tin, plated; 12 percent chromium type corrosion-resistant steels	-0.45	60	
10	Tin-plate; terneplate; tin-lead solder	-0.50	65	
11	Lead, solid or plated; high lead alloys	-0.55	70	
12	Aluminum, wrought alloys of the duralumin type	-0.60	75	
13	Iron, wrought, gray, or malleable; plain carbon and low alloy steels, armco iron	-0.70	85	
14	Aluminum, wrought alloys other than duralumin type; aluminum, case alloys of the silicon type	-0.75	90	
15	Aluminum, cast alloys other than silicon type; cadmium, plated and chromated	-0.80	95	
16	Hot-dip-zinc plate; galvanized steel	-1.05	120	
17	Zinc, wrought; zinc-base die-casting alloys; zinc, plated	-1.10	125	
18	Magnesium and magnesium-base alloys, cast or wrought (most anodic)	-1.60	175	

1/ Compatible couples - potential difference of 0.25 volt maximum between groups.

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6.5.5 Plating. When base metals intended for intermetallic contact form couples not allowed by table VI, they are to be plated with those metals which will reduce the potential difference to that allowed by table VI.

6.6 Supersession data. This specification includes the requirements of MIL-P-13110 (Sig C), Protector, Telephone, which was cancelled 22 June 1966.

6.6.1 Cross reference of military protector types. The following protectors listed under MIL-P-55624 supersede and replace those listed under MIL-P-13110. Existing stock is not affected.

MIL-P-55624	MIL-P-13110	Remarks
M55624/1-001	AR-9-A	For indoor use
M55624/4-001	AR-6-A	For outdoor use

6.6.2 1/ Cross reference of commercial protector types. The following is a cross reference of commercial manufacturers types with MIL-P-13110 types:

Military type designation	Protector		Manufacturer
	Catalog number	Type number	
AR-6-A	1093CW		Western Electric Company
	475-1586	O-52	Cook Electric Company
	1000		Reliable Electric Company
AR-9-A	98A		Western Electric Company
		B-52	Cook Electric Company
	955		Reliable Electric Company

6.6.3 Commercial equivalents.

6.6.3.1 The following commercial fuses are cross-referenced to military types as follows:

Superseding MIL-P-55624 part number	Fuse		Manufacturer	Remarks
	Catalog number	Power rating		
M55624/9-001	11C	7 Amps	Western Electric Co.	Used in protectors AR-6A and AR-9-A under MIL-P-13110
	A-52	7 Amps	Cook Electric Co.	
	55	7 Amps	Reliable Electric Co.	
M55624/10-001	77L	7 Amps	Reliable Electric Co.	Used in protectors M55624/2 M55624/4
	A10U	7 Amps	Cook Electric Co.	

6.6.3.2 The following commercial arrestor assemblies are cross-referenced to military types as follows:

Superseding MIL-P-55624 part number	Commercial arrestor assembly	Manufacturer	Remarks
M55624/5-001	41-225	Cook Electric Company	Used protectors AR-6A and AR-9-A under MIL-P-13110
	P495	Reliable Electric Company	

1/ This information extracted from MIL-P-13110.

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6.6.3.3 The following commercial protectors are cross-referenced to military types as follows:

Superseding MIL-P-55624 part number	Commercial protector	Manufacturer	
M55624/2-001	H2000	Reliable Electric Company	Used in lieu of AR-6-A since cancellation of MIL-P-13110
	H-20-10U	Cook Electric Company	

Custodians:
Army - EL
Navy - YD
Air Force - 80

Review activities:
Army - MU, MI, SL, EL
Navy - EC
Air Force - 80, 11
DSA - ES

User activities:
Army - AT, WC, ME
Navy - MC, SH
Air Force -

Preparing activity:
Army - EL

Agent:
DSA - ES

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